B.Tech.(Computer Engg.) V Semester Examinations

Session 2021-22

Time: 1 hr. **Computer Networks (CEN - 505) Sessional-I** Max Marks: 15

Note: Attempt All Questions

1. **CO1** Suppose a user wants to send a data from a node with logical address 'IP1' and physical address 15, located on one local area network, to a node with a logical address 'IP2' and physical address 85, located on another local area network. If the data coming from the upper layers has service point address 'p1' (Assume 'p1' is the port address of the sending process and 'p2' is the port address of the receiving process at receiver machine) and the size of the data is larger than the network layer can handle. Show the scenario, how data is handled at each layer of OSI model from source to destination. **4**

2. **CO1** An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to four groups of customers as follows: **4**

a. The first group has 32 customers; each needs 512 addresses.
b. The second group has 64 customers; each needs 256 addresses.
c. The third group has 256 customers; each needs 64 addresses
d. The fourth group has 128 customers; each needs 32 addresses.

Design the sub blocks and find out how many addresses are still available after these allocations.

3. **CO2** Differentiate between bit rate and baud rate. What is the baud rate of Biphase Encoding schemes? Encode the following bit stream using B8ZS and HDB3. Assume that the number of 1's so far are odd and the first 1 is positive. **4**

101100000111000000000001010

4. **CO2** Design a TST (Time-Space-Time) switch with 36 inputs and 36 outputs. Assume that 25% of the users are usually active at a time. Calculate minimum number of cross points needed in this scenario. Compare this scenario with the situation when only one cross bar switch is used for all input and output connections. **3**